

# Build a talk like software

manim-presentation-pipeline · 2026-10-01

# An animated talk shows what slides can't.

2



requests

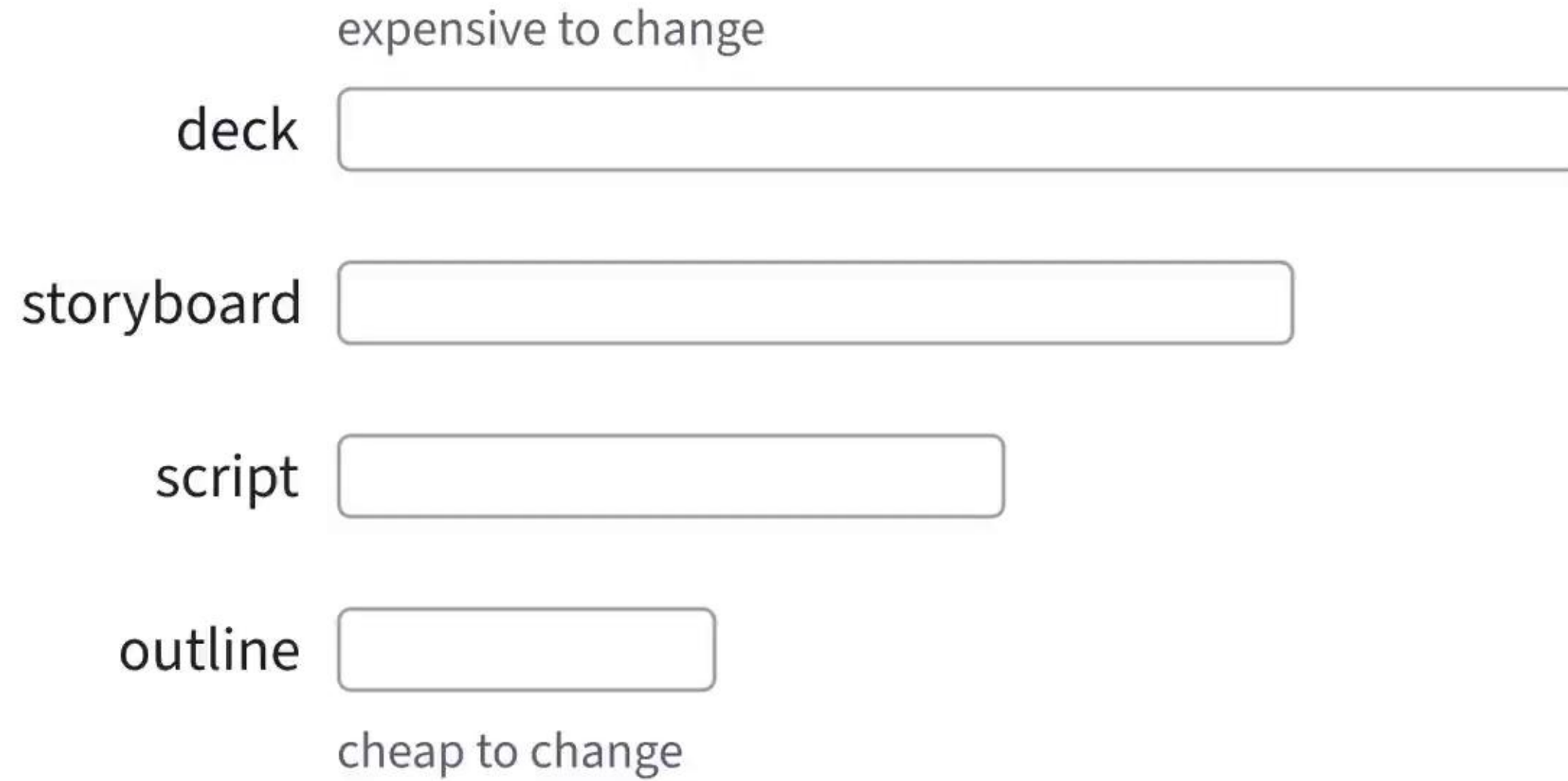


cache

# Every change to an animation is code and a render.

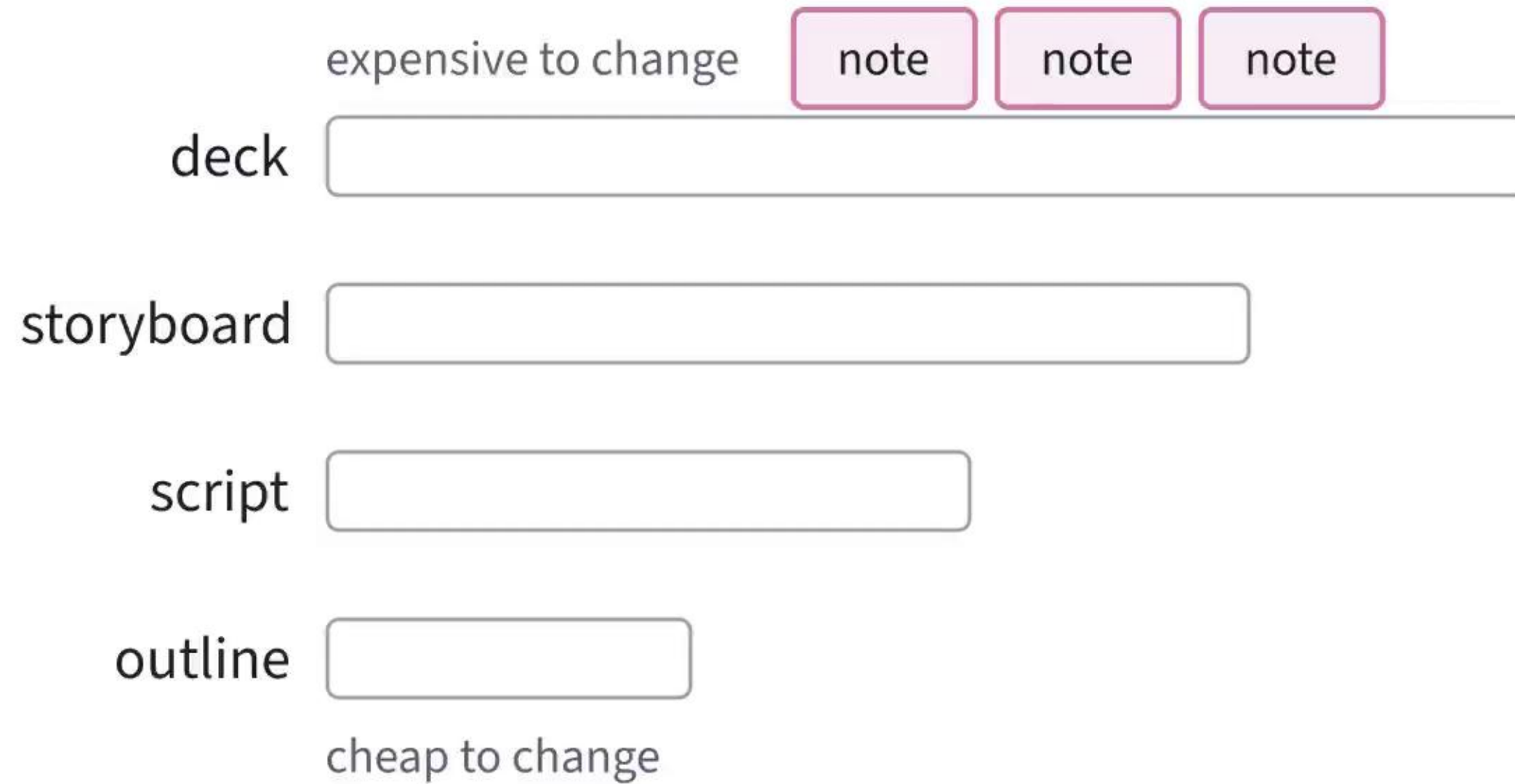
```
class B1(Beat):  
    def construct(self):  
        queue = requests(6)  
        self.click(fill(queue))  
        self.play(warm(cache))
```

# The same note costs more the later it lands.



# Most notes land at the expensive end.

5



Check each decision where it's cheapest to change.

6

brief

digest

outline

script

visual

design

build

rehearsal

final

Nine stages, from a brief to a release.

7

# Warm caches

a talk to the systems group, made with the pipeline

brief

digest

outline

script

visual

design

build

rehearsal

final

# The brief fixes what the talk is for.

|              |                                                     |
|--------------|-----------------------------------------------------|
| brief.md     |                                                     |
| audience     | the systems group                                   |
| slot         | 10 minutes, normal pace                             |
| one sentence | A cache that warms itself halves the slowest waits. |
| held back    | the eviction policy                                 |

brief

digest

outline

script

visual

design

build

rehearsal

final

# The author checks the audience, the slot, the one sentence.

brief.md  
audience    the systems group  
slot        10 minutes, normal pace  
one sentence    A cache that warms itself halves the slowest waits.  
held back    the eviction policy

The author checks

- ☒ who is in the room
- ☒ the slot and the pace
- ☒ the one sentence

brief

digest

outline

script

visual

design

build

rehearsal

final

# The digest lists what the talk could say, with sources.

10

digest.md

claim

source

sort

Most requests repeat within a minute.

logs/

must

The cache fills itself as it serves.

cache.py

could

The slowest waits halve once the cache is warm.

bench/

must

brief

digest

outline

script

visual

design

build

rehearsal

final

# The author sorts every item and answers every gap.

11

digest.md

claim

source

sort

Most requests repeat within a minute.

logs/

must

The cache fills itself as it serves.

cache.py

could

The slowest waits halve once the cache is warm.

bench/

must

The author checks

- ☒ must, could or leave out
- ☒ every gap answered

brief

digest

outline

script

visual

design

build

rehearsal

final

# The outline is the story on one page.

outline.md

| ID | type     | spine line                                      | time |
|----|----------|-------------------------------------------------|------|
| B1 | setup    | At rush hour, users wait.                       | 0:45 |
| B2 | obstacle | A cold cache makes it worse.                    | 0:50 |
| B3 | PEAK     | The slowest waits halve once the cache is warm. | 1:10 |
| B4 | payoff   | The cache warms itself.                         | 0:40 |

# From here on, follow one sentence.

outline.md

| ID | type     | spine line                                      | time |
|----|----------|-------------------------------------------------|------|
| B1 | setup    | At rush hour, users wait.                       | 0:45 |
| B2 | obstacle | A cold cache makes it worse.                    | 0:50 |
| B3 | PEAK     | The slowest waits halve once the cache is warm. | 1:10 |
| B4 | payoff   | The cache warms itself.                         | 0:40 |

# The author checks the picks and the page.

14

outline.md

| ID | type     | spine line                                      | time |
|----|----------|-------------------------------------------------|------|
| B1 | setup    | At rush hour, users wait.                       | 0:45 |
| B2 | obstacle | A cold cache makes it worse.                    | 0:50 |
| B3 | PEAK     | The slowest waits halve once the cache is warm. | 1:10 |
| B4 | payoff   | The cache warms itself.                         | 0:40 |

The author checks

- ☒ a pick per decision
- ☒ the story on one page



# The script is what the speaker says, written for the ear.

15

### B3. The cache warms - 1:10

**The slowest waits halve once the cache is warm.**

> the bars of the slow requests shrink

brief



digest



outline



script

visual

design

build

rehearsal

final

# The author reads every act aloud.

16

### B3. The cache warms - 1:10

**The slowest waits halve once the cache is warm.**

> the bars of the slow requests shrink

The author checks

- ☒ every act read aloud
- ☒ the total fits the slot

brief



digest



outline



script



visual

design

build

rehearsal

final

# The storyboard splits the script into clicks.

17

B3.2

```
[[B3.frame]]
```

```
head = "The slowest waits halve once the cache is warm."
```

```
trigger = "once the cache is warm"
```

```
narration = "..."
```

brief



digest



outline



script



visual

design

build

rehearsal

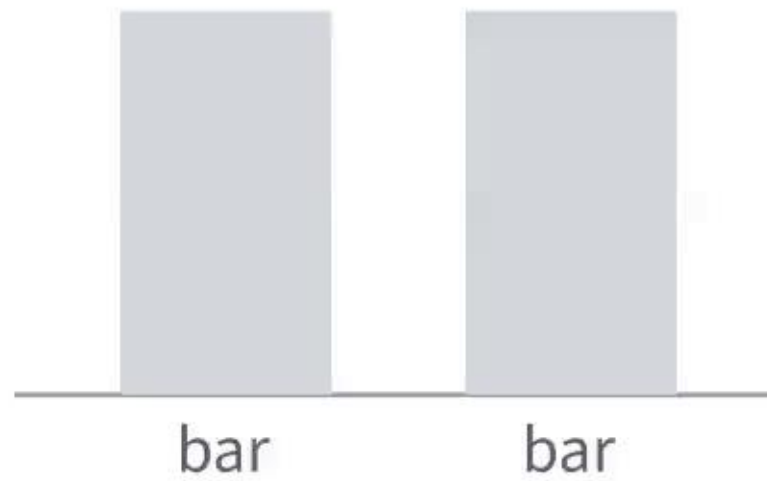
final

# The greybox shows every slide as rough boxes.

18

The slowest waits halve once the cache is warm.

7



brief



digest



outline



script



visual

design

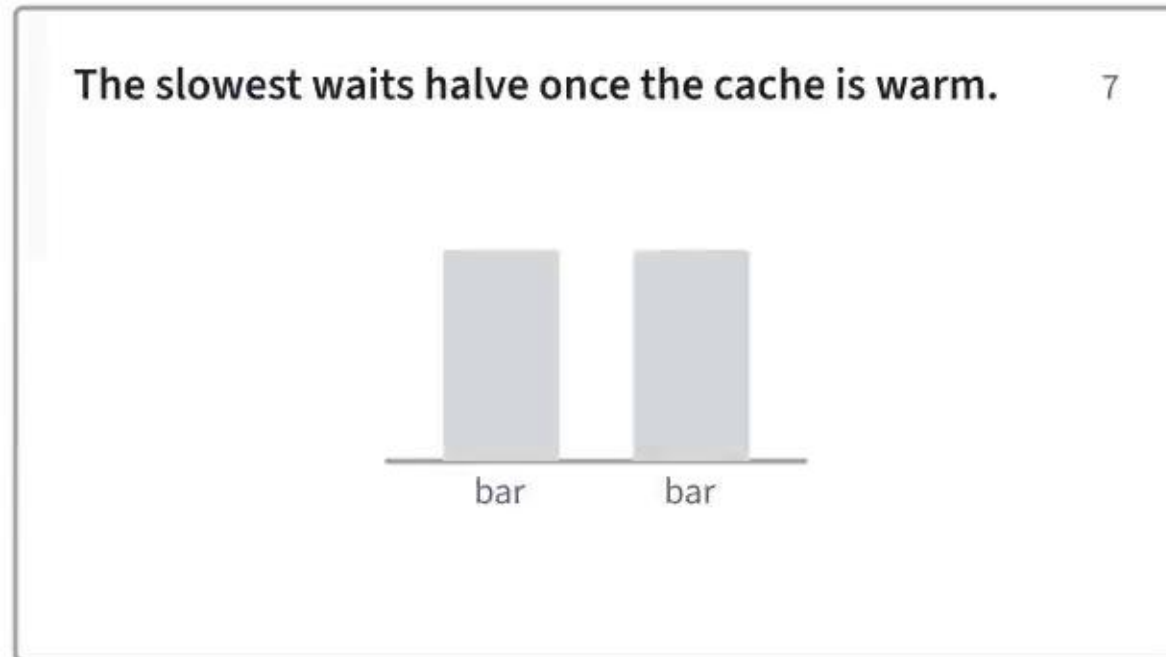
build

rehearsal

final

# The author checks the greybox, slide by slide.

19



The author checks

- ☒ a pick per beat
- ☒ one claim per slide
- ☒ each click on its word



# The design system fixes the look once.

20



# The author checks the specimen.

21



The author checks

☒ the specimen, once

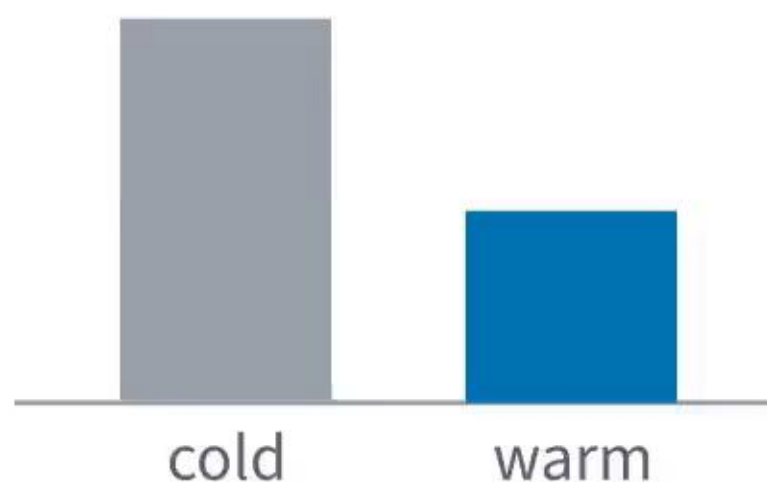


# The build renders the deck, drafts first.

22

The slowest waits halve once the cache is warm.

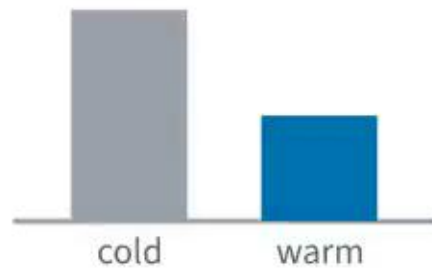
7



# The author checks stills, then clicks through.

23

The slowest waits halve once the cache is warm. 7



The author checks

- ☒ stills of every slide
- ☒ a click-through per act



# Rehearsal: a late note goes back to the earliest stage it changes.

slide 7: say halves, not 48%



Slide 7 is beat 3, frame 2 in the draft the reviewer saw.

25

slide 7: say halves, not 48%

slide 7 = B3.2, in that draft



# The note is said back with its stage, and fixed there.

26

stage: script

slide 7: say halves, not 48%

slide 7 = B3.2, in that draft



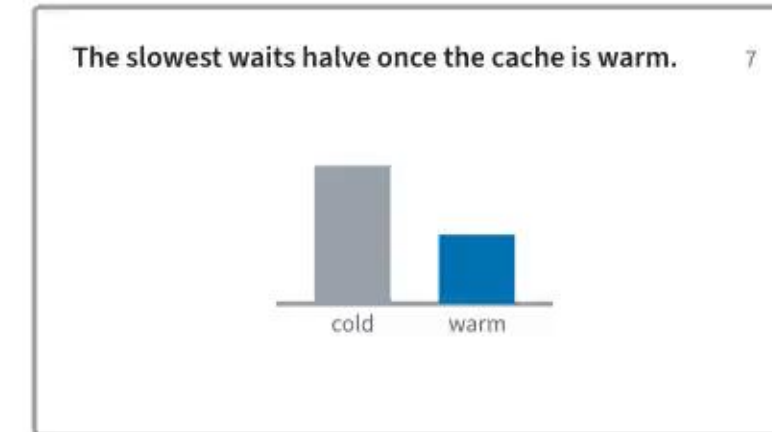
# Everything after it is rebuilt; only what changed is shown again.

27

stage: script

slide 7: say halves, not 48%

slide 7 = B3.2, in that draft



# The final stage releases the talk.

28

releases/global/v1/

[talk.html](#)

the animated deck, offline

[talk.pdf](#)

one page per click

[script.pdf](#)

the script, with every click

[design.md](#), [tokens.toml](#)

the design system



# The author clicks through once, end to end.

29

releases/global/v1/

[talk.html](#)

the animated deck, offline

[talk.pdf](#)

one page per click

[script.pdf](#)

the script, with every click

[design.md](#), [tokens.toml](#)

the design system

The author checks

☒ one click-through, end to end

brief

digest

outline

script

visual

design

build

rehearsal

final

Clone it, write only in talk/  
point any agent at AGENTS.md.

AGENTS.md  
pipeline/  
engine/  
talk/  
    source/  
    global/  
    variants/  
vendor/

### AGENTS.md

read the principles, run mpp status,  
follow the stage it names



One command checks the machine;  
one builds the deck, offline.

AGENTS.md  
pipeline/  
engine/  
talk/  
    source/  
    global/  
    variants/  
vendor/

### AGENTS.md

read the principles, run mpp status,  
follow the stage it names

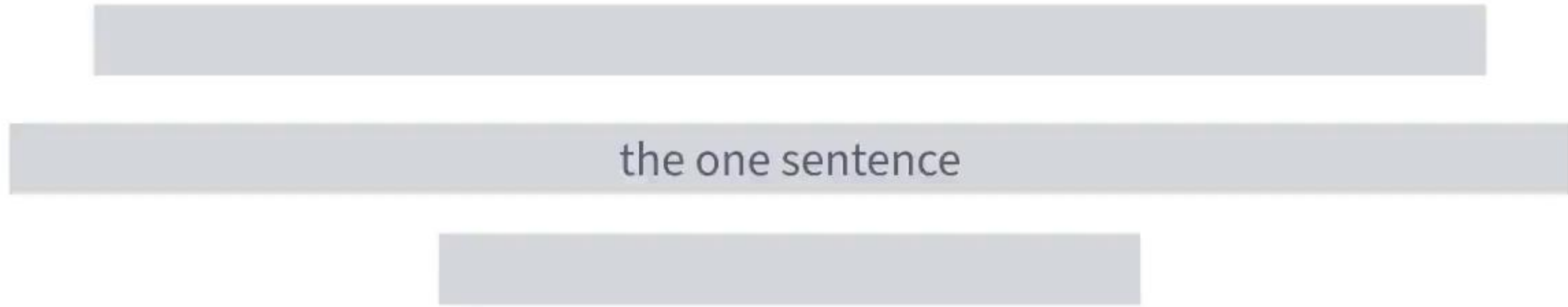
`uv run mpp doctor` checks the machine

`uv run mpp build` builds the deck, offline



# This slide started as a grey box.

32



Every stage was checked before the next.

33

**Build a talk the way you build software:**  
in stages, each checked cheaply by its author,  
from one source file.



# The build checks every click for text over text.

34



two labels on top of each other, found before any render

# And every seam between slides for a jump.

35



they differ: the audience sees a jump